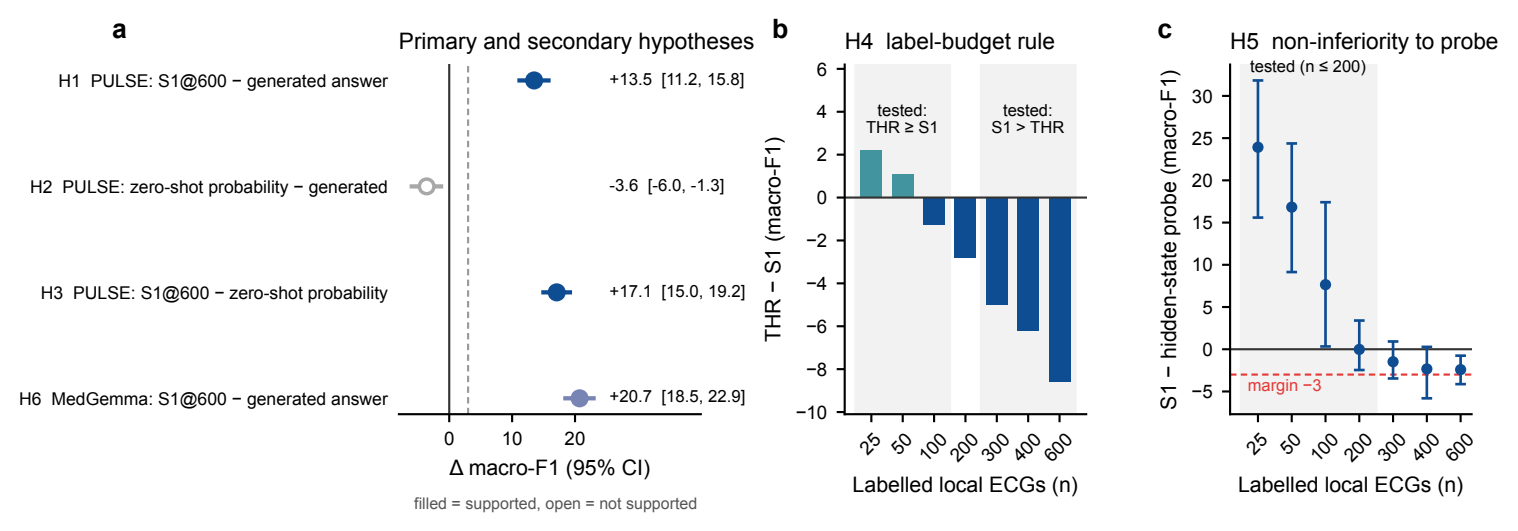
a
Primary and secondary hypotheses
H1 PULSE: S1@600 − generated answer
+13.5 [11.2, 15.8]
H2 PULSE: zero-shot probability − generated
-3.6 [-6.0, -1.3]
H3 PULSE: S1@600 − zero-shot probability
+17.1 [15.0, 19.2]
H6 MedGemma: S1@600 − generated answer
+20.7 [18.5, 22.9]
0
10
20
Δ macro-F1 (95% CI)
filled = supported, open = not supported
b
H4 label-budget rule
tested:
THR ≥ S1
tested:
S1 > THR
THR − S1 (macro-F1)
6
4
2
0
−2
−4
−6
−8
−10
25
50
100
200
300
400
600
Labelled local ECGs (n)
c
H5 non-inferiority to probe
tested (n ≤ 200)
S1 − hidden-state probe (macro-F1)
30
25
20
15
10
5
0
−5
margin −3
25
50
100
200
300
400
600
Labelled local ECGs (n)